



LABORATORY INSTRUCTION

Control Engineering Laboratory

Modelling and Control of DC Motor Speed and Position

BACHELOR - MGST

last update: May 11, 2026

Contents

1	Learning Target	1
2	Hardware Overview	2
2.1	Hardware	2
2.1.1	Mechanical implementation	2
2.1.2	Motor driver	3
2.1.3	Quadrature encoder	3
2.1.4	Electrical implementation	3
3	Preparatory Work	4
3.1	Analytical modeling of a DC motor	4
3.2	System identification	4
4	Lab Tasks	6
4.1	Setup hardware	6
4.2	Setup software	6
4.3	Identification of the system	6
4.4	Controller design	8
4.5	Controller implementation and evaluation	9
5	Report	10
	List of Figures	III
	Bibliography	IV
6	Appendix	IV

1 Learning Target

A DC motor is a type of electro-mechanical actuator widely used in automotive, medical devices, industrial automation, entertainment, home appliances, and aerospace.

The learning goal for this lab is an identification of an electro-mechanical system based on measurement of the step response and the design of a suitable controller by employing MATLAB Simulink and National Instrument data acquisition device (NI DAQ). In the course of the lab the measured signal has to be processed for further use and approximated with a standard element. The following steps are to be done in the laboratory:

- Measurement of step response
- System identification
- Design of a PID controller
- Simulation
- Implementation

This laboratory work applies knowledge from the courses “Control Engineering ” in combination with “Analog circuit” and “Sensor and Actuators”.

2 Hardware Overview

2.1 Hardware

2.1.1 Mechanical implementation

The hardware includes a gear box and a DC motor (Faulhaber 3257G012CR, or Maxon RE35 273752). The datasheet of corresponding motor can be found in Appendix 6.

The quantity of interest to be controlled is the angular position of the pointer attached to the end step. Angular position is measured directly on the output shaft by an optical quadrature encoder. The transmission takes place via a belt drive. Figure 1 depicts the system with its components labelled.

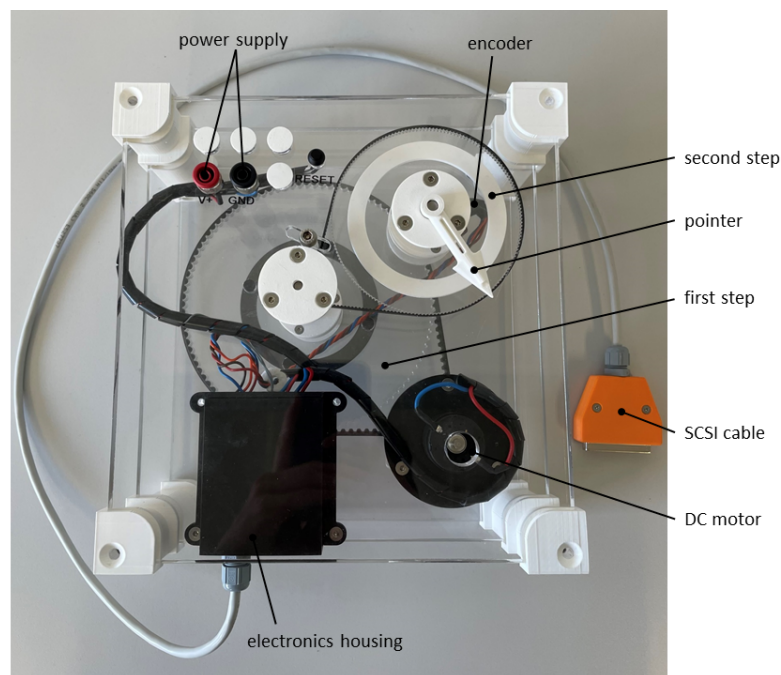


Figure 1: Mechanical belt transmission system driven by a DC motor.

Warning: Avoid touching any moving part of the hardware under any circumstance, as doing so could result in severe injuries!

Note: For the laboratory four different boxes with different transmission ratios i have been implemented. The transmission ratios of each box are listed below.

- Box 1 $\rightarrow i = 97.2$
- Box 2 $\rightarrow i = 73.5$
- Box 3 $\rightarrow i = 103.3$
- Box 4 $\rightarrow i = 103.7$

2.1.2 Motor driver

The motor driver employed for this laboratory is the Cytron MD13S, enabling the control of a motor speed through a low-power pulse-width modulated (PWM) signal derived from the digital output of the National Instruments (NI) PCIe-6321 DAQ card.

2.1.3 Quadrature encoder

The optical encoder utilized in this lab is the Broadcom Avago HEDS-5500 A11. It functions as an incremental encoder, optically sensing position changes with a resolution of 500 counts per revolution, determining the shaft's angular position. By accounting for the sampling time and position change between two samples, it can calculate the pointer's angular velocity. Additionally, the quadrature encoder offers two channels to ascertain the direction of rotation. The datasheet has been uploaded to the designated resource folder on Sakai.

2.1.4 Electrical implementation

The electronics housing contains a motor driver and a circuit board, linking the I/O pins of the BNC-2090a module to corresponding pins of the encoder and motor driver. Meanwhile, the BNC-2090a is a front panel connected to the PCIe-6321 DAQ card, which controls the hardware. A detailed schematic illustrating the interconnection of individual electrical components is provided in Figure 2.

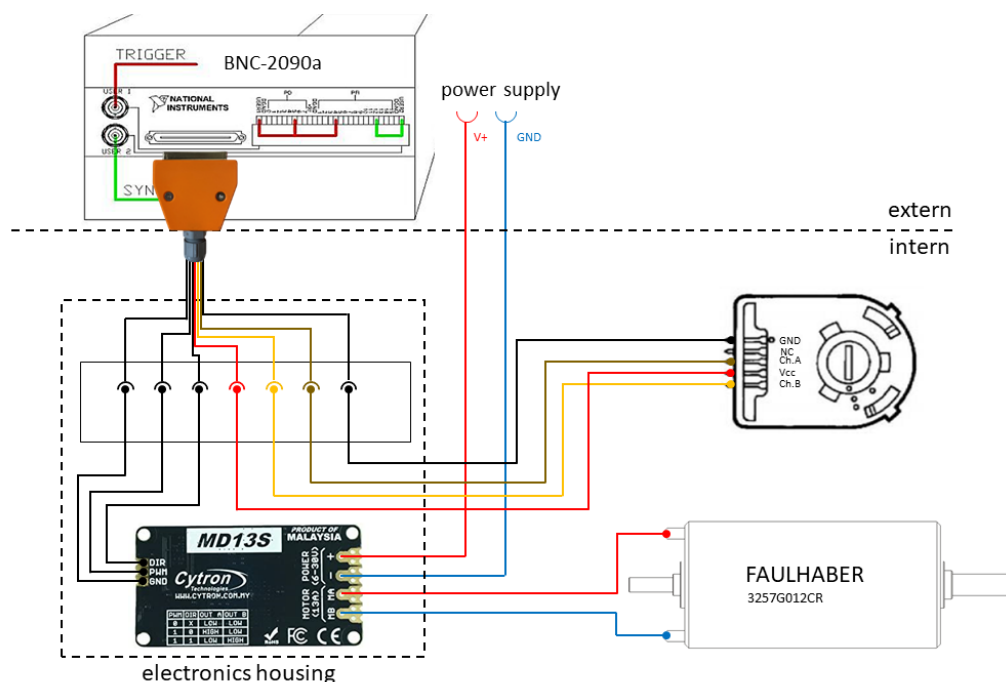


Figure 2: Schematic of wiring of electrical components.

3 Preparatory Work

Please complete the following tasks in preparation. Bring your preparatory work to the lab session as you will need it to identify and control the actual system during the session.

3.1 Analytical modeling of a DC motor

Figure 3 shows a simplified schematic representation of a DC motor. In this example, the motor is considered unloaded, resulting in zero values for J_{load} and b_{load} . Conduct white box modeling approach to determine the transfer function $G_p(s)$ for a DC motor. Please derive two transfer functions:

- $G_{p1}(s)$: Voltage - input, Angular displacement - output
- $G_{p2}(s)$: Voltage - input, Angular velocity - output

Determine a state-space model for the motor when the voltage is the input and the angular displacement is the output. Hint: You can select the state variables as angular displacement, angular velocity, and current.

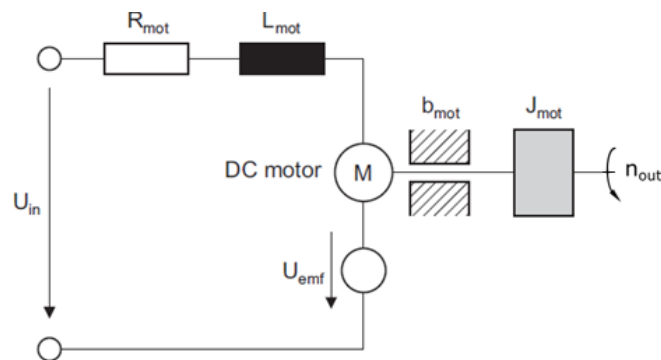


Figure 3: Schematic representation of the electro-mechanical system.

3.2 System identification

Bring Exercise 4-2 solutions or complete the following task. The *measurement_data.mat* file located in the Sakai folder contains measurement data of the step response of an arbitrary mechatronic system. The time in seconds is recorded in the first column. The second column contains the system input values and the system output values were recorded in the third column, both in volts. Be aware that a magnitude of the step input is not always 1.

- Plot the raw measurement data similar to Figure 4.
- Identify the system from the raw data to a PT2 over-damped system.

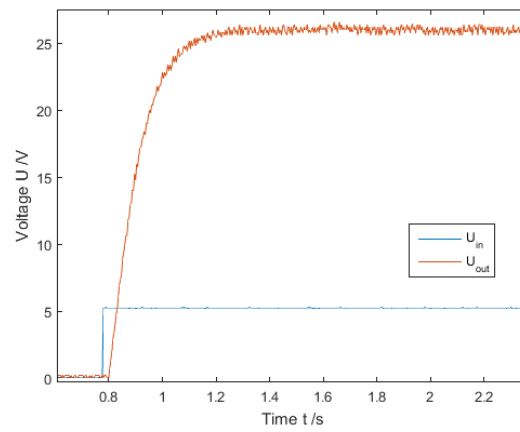


Figure 4: Raw measurement data of the step response

4 Lab Tasks

4.1 Setup hardware

A gear box system actuated by a DC motor and controlled by the NI PCIe-6321 DAQ card are used for this laboratory. To setup the hardware for the experiment the following steps must be carried out:

1. Connect the voltage output of the power supply to the device's corresponding labeled plugs.
2. Connect the SCSI cable from the device's electronics housing to the NI BNC-2090a I/O module.

4.2 Setup software

The main part of the laboratory is carried out on the Desktop PC located in the MCI VI laboratory rooms. These laboratory PCs are equipped with NI PCIe-6321/6221 DAQ cards, which enable real-time transmission, acquisition and analysis of analog and digital data. MATLAB Simulink is used for system identification, simulation of the model and implementation of the control system. Therefore, it utilizes the Simulink Desktop Real-Time add-On to communicate with the NI PCIe-6321/6221 DAQ card. This provides an easy and comprehensible visualization of control systems while keeping the coding effort substantially low.

Even though all relevant data and software for this lab are pre-installed and set up, it is advisable to follow the checklist provided below in case any components behave unexpectedly.

- Start MATLAB and open and run the .m-file named *"SystemID_Setup"*
- Go to the *"Desktop Real-Time"* panel and click on *"Hardware settings"*.
 - Hint: if the option is not visible right away, extend the drop-down menu in the *"Apps"* register and look for the option there.
- First, go to the *"Solver"* register and:
 - Set start time to 0 and end time to 1.5.
 - Under *"Solver settings"* set type to *"Fixed-step"*, expand *"Solver details"* and set *"Fixed-step size"* to *0.0001s*.
- Next, click on the *"Code Generation"* register, and browse *"sldrt.tlc"* as the *"System target file"*

4.3 Identification of the system

Open the Simulink model *"EncoderRead.slx"* (see Figure 5) via execution of the MATLAB script *"SystemID_Setup.m"*.

Your task is to record the system's step response for an input step magnitude of 12 V and without any delay. Connect the provided elements appropriately to accomplish this.

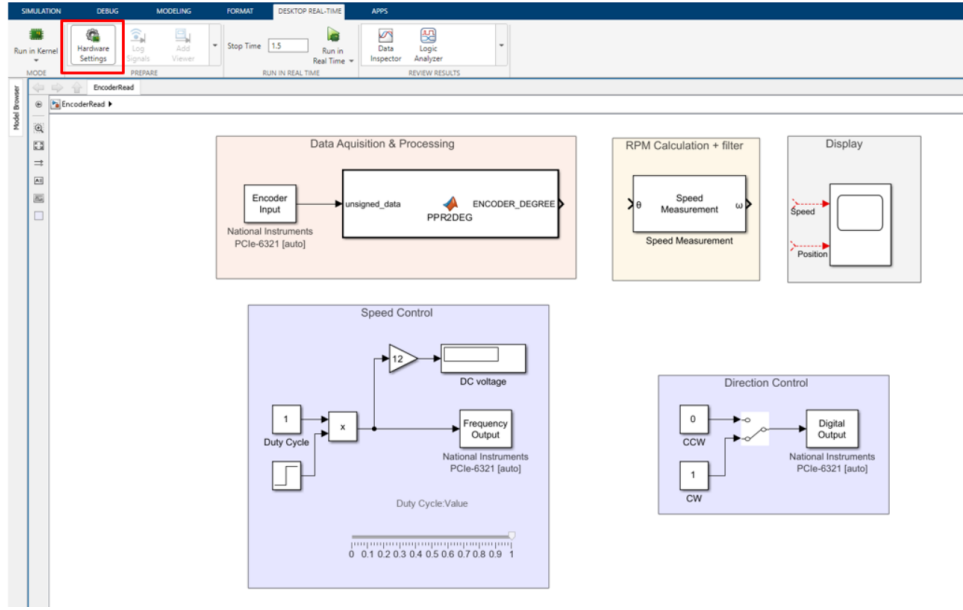


Figure 5: Simulink interface for system identification

Note: Please note that the scope block measures the angular velocity, ω in revolutions per minute (rpm) and angular position θ in degrees. The measured data is logged to the MATLAB workspace upon completion of the measurement process.

After measuring the step response of the system, determine two transfer functions of the system with the DC motor by applying a least-square fitting method.

Note: Consider the unit of the measured data based on which the system is modelled. To be able to control the box with desired values being angular position in degrees, a conversion from rpm to $\frac{\text{rad}}{\text{s}}$ must be conducted.

The general form of PT2 system transfer function is given in Equation 1.

$$\text{PT2}(s) = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)} \quad (1)$$

Plot the step response of the identified transfer function for an input voltage of 12 V and compare it to the measured step response. What is the output variable for this 2nd order system? Your results should look somewhat similar to Figure 6.

Since we intend to control the position θ of the arrow, we need to adapt the estimated transfer function $G_p(s)$ of the model. This must be done since the measured data describes the arrow's angular velocity ω for a certain input voltage u . Their relation is given in Equation 2.

$$\text{PT3}(s) = G_p(s) = \frac{\Theta(s)}{U(s)} = \frac{\Omega(s)}{s U(s)} = \frac{1}{s} \text{PT2}(s) \quad (2)$$

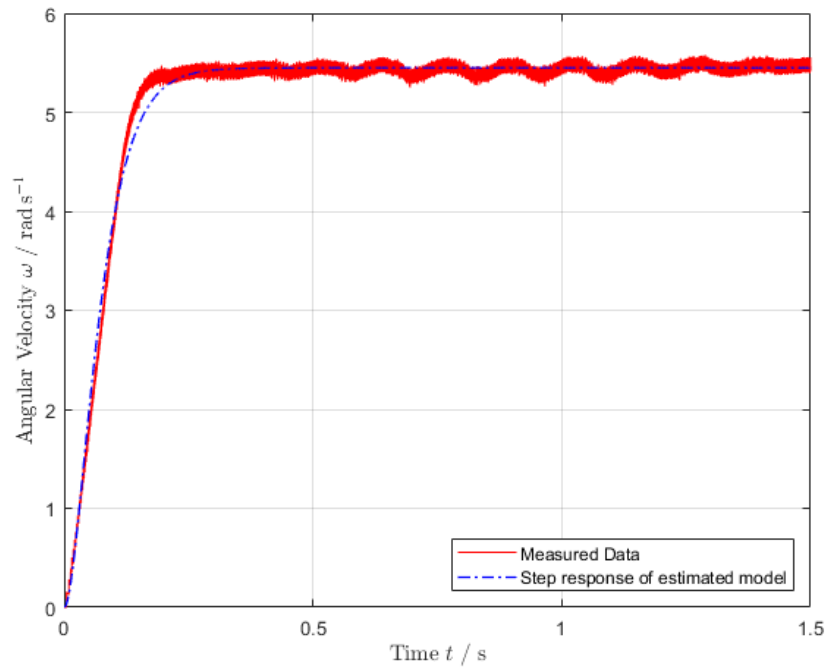


Figure 6: Example of raw measurement data of the step response

4.4 Controller design

Design a PID-controller for a closed-loop control system by applying the Ziegler-Nichols method for the identified system and simulate the closed-loop system.

Note 1: For method 1 the requested parameters can be read from the step response of the system.

Note 2: In case of method 2 you must use the Matlab function `"rlocus()"` based on your system transfer function to find the critical gain K_{cr} and design another closed-loop control system with P-Controller in a blank Simulink model, as depicted in Figure 7, to yield the period time P_{cr} from the output signal.

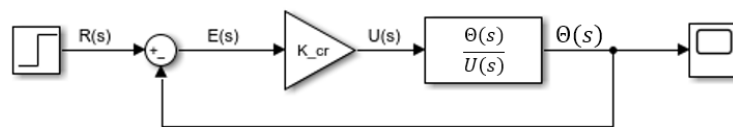


Figure 7: Block diagram of closed-loop control system with P-controller

After acquiring a model of the system and the PIC controller, we can simulate the performance of a closed-loop control system and measure the input $R(s)$, error $E(s)$, control

$U(s)$ and output $\Theta(s)$ signals by using the scope block. Please include R2D (radians to degrees), D2R (degrees to radians) block as shown in Figure 8.

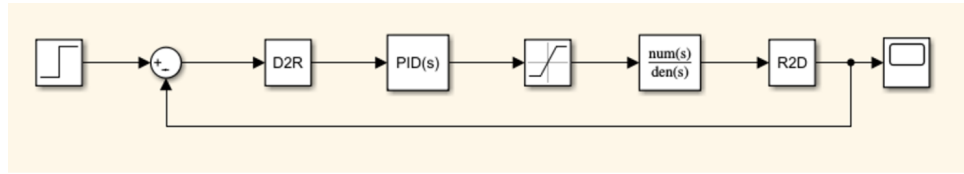


Figure 8: Block diagram of closed-loop control system with P-controller

4.5 Controller implementation and evaluation

After simulating the designed closed-loop control system with a PID controller, proceed to implement the control system using Simulink Desktop Real-Time. Open the MATLAB file named "RunController.m", input the PID gain coefficients as directed in the code, and execute the file. The block diagram you are presented with is incomplete. Therefore complete the block diagram and apply for Anti-windup method (e.g. clamping) in your PID controller.

Observe the behaviour of the actual system and answer the following questions:

- Does it behave as expected? If not, where do your observations deviate from your expectations?
- Tune K_p , K_i , and K_d gains and then describe how these adjustments influence the performance of the position control.

5 Report

Prepare a report with **maximum of 10 pages**, providing comprehensive explanations for all preparatory work and lab tasks through calculations and drawings. Please make sure to document your reasoning, approach, results, and interpretations in a scientific manner.

Consider the following aspects when preparing the report:

- The report should follow to the MCI Guidelines for documenting laboratory activities. Please keep it as concise as possible while ensuring it contains all relevant information.
- The data collected during laboratory activities should be analyzed with detailed commentary on measurement outcomes, comparing them with theoretical expectations and simulations.
- At the the report, there should be a section for drawing conclusions and discussions.

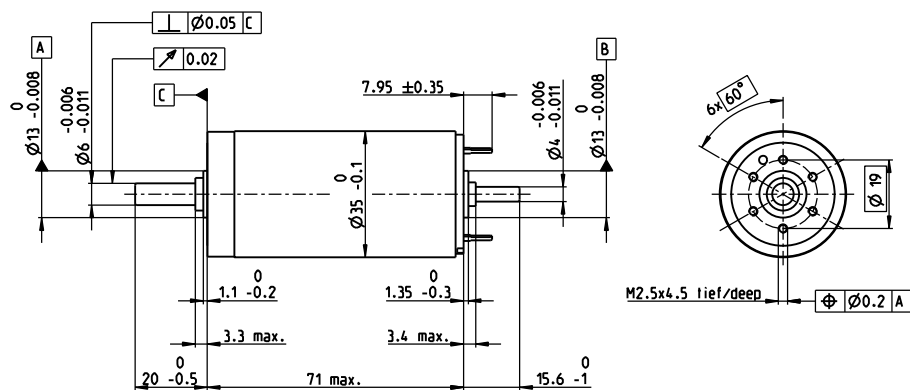
The report should be submitted on Sakai - Assignments by the deadline. Please include the lab date, group members, and other important information on the cover page.

List of Figures

1	Mechanical belt transmission system driven by a DC motor.	2
2	Schematic of wiring of electrical components.	3
3	Schematic representation of the electro-mechanical system.	4
4	Raw measurement data of the step response	5
5	Simulink interface for system identification	7
6	Example of raw measurement data of the step response	8
7	Block diagram of closed-loop control system with P-controller	8
8	Block diagram of closed-loop control system with P-controller	9

6 Appendix

RE



Lagerprogramm
 Standardprogramm
 Sonderprogramm (auf Anfrage)

gemäss Massbild
Wellenlänge 15.6 gekürzt auf 4 mm

273752	323890	273753	273754	273755	273756	273757	273758	273759	273760	273761	273762	273763
285785	323891	285786	285787	285788	285789	285790	285791	285792	285793	285794	285795	285796

maxon DC motor 139

DC-Kleinstmotoren

Graphitkommutierung

73 mNm
60 W

Serie 3257 ... CR

Werte bei 22°C und Nennspannung		3257 G	012 CR	018 CR	024 CR	036 CR	048 CR	
1	Nennspannung	U_N	12	18	24	36	48	V
2	Anschlusswiderstand	R	0,41	0,84	1,63	4,15	6,56	Ω
3	Wirkungsgrad, max.	η_{max}	83	83	83	80	83	%
4	Leerlaufdrehzahl	n_0	5 700	6 100	5 900	5 600	5 900	min ⁻¹
5	Leerlaufstrom, typ. (bei Wellen ø 5 mm)	I_0	0,258	0,18	0,129	0,082	0,064	A
6	Anhaltemoment	M_H	531	561	539	518	547	mNm
7	Reibungsdrehmoment	M_R	4,9	4,9	4,9	4,9	4,9	mNm
8	Drehzahlkonstante	k_n	500	352	253	156	125	min ⁻¹ /V
9	Generator-Spannungskonstante	k_E	2	2,84	3,95	6,4	7,98	mV/min ⁻¹
10	Drehmomentkonstante	k_M	19,1	27,2	37,7	61,1	76,2	mNm/A
11	Stromkonstante	k_I	0,052	0,037	0,027	0,016	0,013	A/mNm
12	Steigung der n-M-Kennlinie	$\Delta n / \Delta M$	10,7	10,9	10,9	10,6	10,8	min ⁻¹ /mNm
13	Anschlussinduktivität	L	70	140	270	700	1 100	μ H
14	Mechanische Anlaufzeitkonstante	τ_m	4,7	4,7	4,7	4,7	4,7	ms
15	Rotorträgheitsmoment	J	42	41	41	42	42	gcm ²
16	Winkelbeschleunigung	α_{max}	130	140	130	120	130	·10 ³ rad/s ²
17	Wärmewiderstände	R_{th1} / R_{th2}	2 / 8					K/W
18	Thermische Zeitkonstante	τ_{w1} / τ_{w2}	17 / 810					s
19	Betriebstemperaturbereich:							
	– Motor		-30 ... +125					°C
	– Wicklung, max. zulässig		+155					°C
20	Wellenlagerung		Kugellager, vorgespannt					
21	Wellenbelastung, max. zulässig:							
	– für Wellendurchmesser		5					mm
	– radial bei 3 000 min ⁻¹ (3 mm vom Lager)		50					N
	– axial bei 3 000 min ⁻¹		5					N
	– axial im Stillstand		50					N
22	Wellenspiel:							
	– radial	≤	0,015					mm
	– axial	=	0					mm
23	Gehäusematerial		Stahl, schwarz beschichtet					
24	Masse		242					g
25	Drehrichtung		rechtsdrehend auf Abtriebswelle gesehen					
26	Drehzahl bis	n_{max}	7 000					min ⁻¹
27	Polpaarzahl		1					
28	Magnetmaterial		NdFeB					

Hinweis: Nennwerte gelten für Nennspannung bei Umgebungstemperatur 22°C und Reduktion des Wärmewiderstandes R_{th2} um 25%.

Hinweis:

Angegeben ist der Bereich der möglichen Arbeitspunkte der Antriebe bei einer Umgebungstemperatur von 22°C.

Das Diagramm beschreibt die empfohlenen Drehzahlbereiche in Abhängigkeit vom Wellendrehmoment. Die Darstellung beinhaltet sowohl den Betrieb im thermisch isolierten als auch im gekühlten Zustand (R_{th2} um 50% reduziert).

Die Nennspannungskurve beschreibt die Betriebspunkte bei U_N im ungekühlten und gekühlten Zustand. Betriebspunkte oberhalb dieser Kurven benötigen eine Versorgungsspannung $> U_N$, Betriebspunkte unterhalb dieser Kurven $< U_N$.

